Temperature and the multiplier in ngspice

How m, temp, dtemp and dt reach an OSDI (Verilog-A) device – routing, conventions, and verification

Ngspice + OpenVAF Enhancements project

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Contents

Temperature and the multiplier in ngspice: **m**, **temp**, **dtemp**, **dt**

How the four instance knobs ngspice supplies on top of a compact model reach an OSDI (Verilog-A) device — and what changes when the model declares them itself.

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1. What these four are

Every SPICE device, built-in or compiled from Verilog-A, sits under four knobs that belong to the simulator, not to the model:

knob	meaning	units on the netlist
`m`	device multiplicity — *n* identical devices in parallel	dimensionless
`temp`	absolute device temperature, overriding the ambient	°C
`dtemp`	temperature offset added to the ambient	ΔK (= Δ°C)
`dt`	the raw OSDI spelling of `dtemp`	ΔK

They are not Verilog-A language features. A model never has to declare them — ngspice provides them for *every* OSDI device, and a model reads the result through the standard system functions `$mfactor`, `$temperature` and `$vt`.

The point of this note is that all four have a routing rule that is easy to misread from either side, and two of them were genuinely broken until Enhancement-394.

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2. They are supplied by default, not on request

This is the first thing to get right, because the intuition runs the other way.

In `osdiregistry.c`, `dt` and `temp` are initialised to synthetic parameter ids *before* the model's own parameters are scanned:

```
uint32_t dt   = descr->num_params + descr->num_opvars;
bool     has_m = false;
uint32_t temp = descr->num_params + descr->num_opvars + 1;
```

and `osdiinit.c` registers them whenever those ids are still valid:

```
if (entry->dt   != UINT32_MAX) { /* register "dt" and "dtemp" */ }
if (entry->temp != UINT32_MAX) { /* register "temp"           */ }
```

So the default is present. Scanning the model's parameters can only take them away or redirect them:

the Verilog-A declares	what ngspice does
(nothing)	provides `temp`, `dtemp`, `dt` and the `m` alias itself
`m`	`has_m = true` → registers no `m` alias; the model owns the name
`temp`	`temp = UINT32_MAX` → registers no `temp`
`dt`	`dt = UINT32_MAX` → registers neither `dt` nor `dtemp`
`dtemp`	`dt = param_id` → the loader's entries are routed to the model's parameter
`temperature`	`temp = param_id` → `temp` routed to the model's parameter

A probe module that declares none of them still answers to all four:

```
`include "disciplines.vams"
module probe(p,n);
 inout p,n; electrical p,n;
 (* desc="tdev" *) real tdev;
 analog begin
   tdev = $temperature;
   I(p,n) <+ V(p,n)*1e-3;
 end
endmodule
```

```
N1 a 0 mm temp=75     * -> $temperature = 348.15 K
N1 a 0 mm dtemp=10    * -> $temperature = 310.15 K
N1 a 0 mm dt=10       * -> $temperature = 310.15 K
N1 a 0 mm m=3         * -> 3x the current
```

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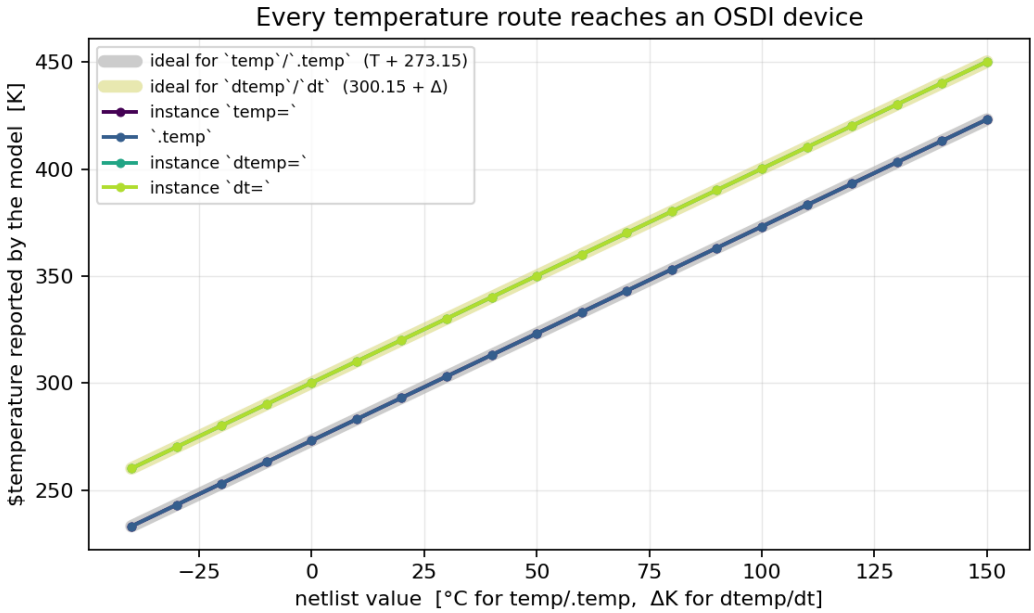
3. Temperature: every route, and the Celsius convention

`temp` is written in degrees Celsius and the device works in Kelvin. Every built-in adds `CONSTCtoK` in its own parameter setter — `dioparam.c` does `DIOtemp = value->rValue + CONSTCtoK`. The OSDI path did not: it stored the raw number and used it directly as the Kelvin device temperature, so `temp=75` reached the model as `$temperature = 75` and `$vt = 6.5 mV` instead of 30 mV. On a Verilog-A diode

that is -2.5×10^{16} A where the correct answer is -4.85×10^{-7} A, and `temp=0` made `$vt` exactly zero so `limexp(V/$vt)` divided by zero. Enhancement-394 fixed it; §7 shows the check that keeps it fixed.

The four routes and where they land:



The two grey reference bands are the closed forms — `T + 273.15` for the absolute knobs, `300.15 + Δ` for the relative ones. `temp` and `.temp` coincide exactly; `dtemp` and `dt` are the same knob under two spellings and coincide with each other.

Rules that follow

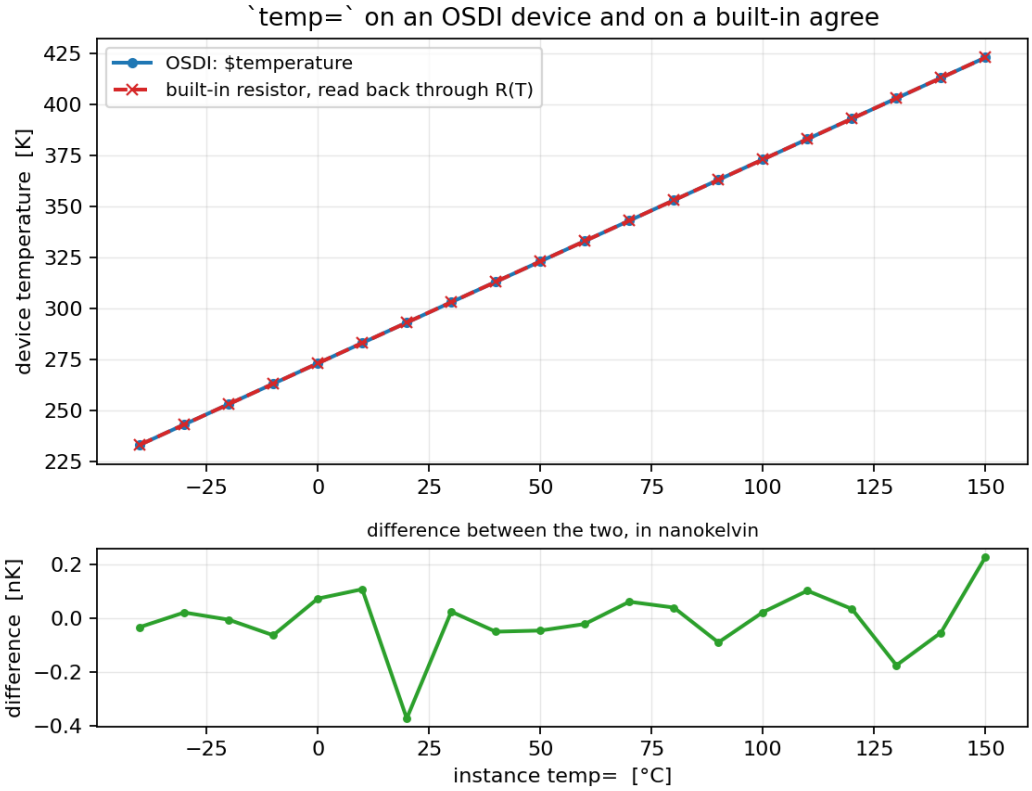
- **`temp=`** overrides **`dtemp=`**, and says so: `n1: Instance temperature specified, dtemp ignored`. (A built-in resistor in the same situation prints nothing, so the OSDI path is the more informative of the two here.)
- **`.temp`** and **`.option temp`** set the ambient, which `dtemp`/`dt` are then relative to: `.temp 75` with `dtemp=10` gives 358.15 K.
- `dt` and `dtemp` are the same parameter. They share an id; writing both is writing one slot twice.

Against an independent thermometer

"The OSDI number moved" is not evidence. The check that matters is whether a built-in device in the same deck sees the *same* temperature. A resistor with a first-order tempco is a thermometer — invert

$$R(T) = R_0\left[1 + \mathrm{tc1}\,(T - T_{\mathrm{nom}})\right]$$

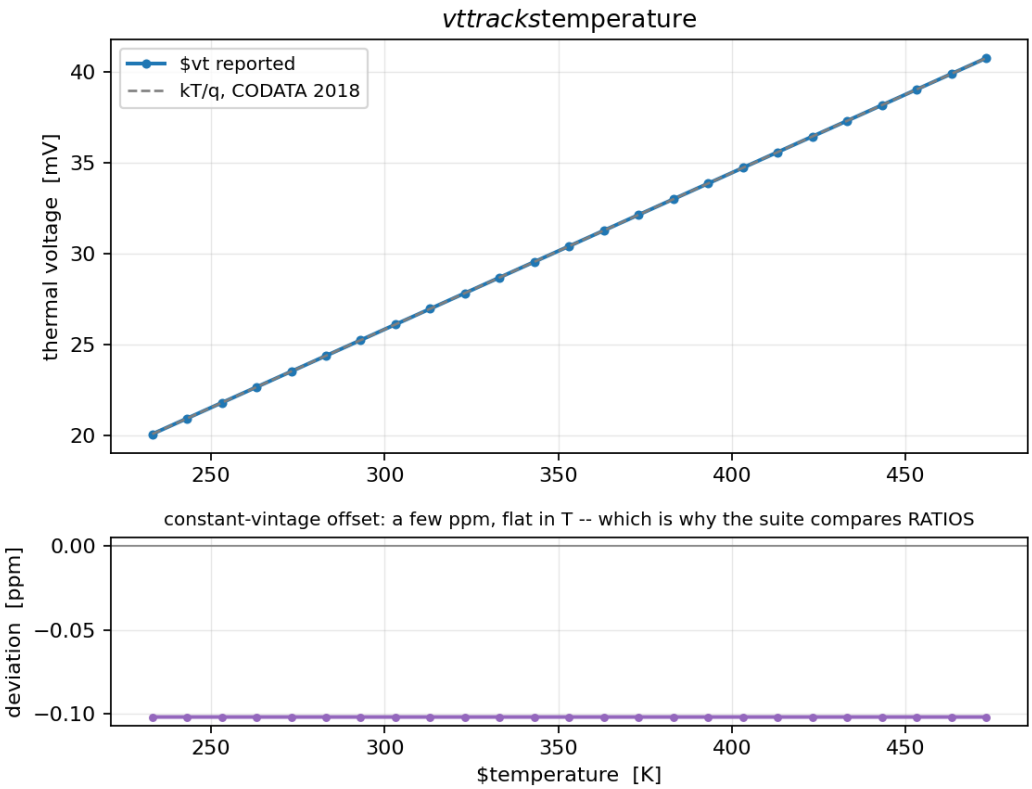
for `T` from the measured current and compare:



The lower panel is the difference, in nanokelvin — it is round-off in the resistor read-back, not a discrepancy.

`$vt` and the constants vintage

`$vt` tracks `$temperature` exactly, but it does not equal a textbook `kT/q` to the last digit:



`constants.vams` and ngspice's own `CONSTboltz`/`CHARGE` are different CODATA vintages, so there is a few-ppm offset that is flat in temperature. That is not a defect, and it is the reason the regression suite checks `$vt` by a ratio — `vt(T₂)/vt(T₁) == T₂/T₁`, which holds to 1 part in 10^{12} — rather than against an absolute constant. An absolute comparison would either fail spuriously or need a tolerance loose enough to hide a real error.

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4. The multiplier, and why `m` is not `$mfactor`

`$mfactor` is a parameter OpenVAF always emits. ngspice exposes it under the netlist name `_mfactor` (the `$`→`_` rewrite in `osdiinit.c`) and normally registers an extra alias keyword `m` pointing at the same id:

```
if (!has_m && !strcmp(para->name[0], "$mfactor")) {
    (*dst)[num_names] = (IFparm){.keyword = "m", .id = (int)i, ...};
}
```

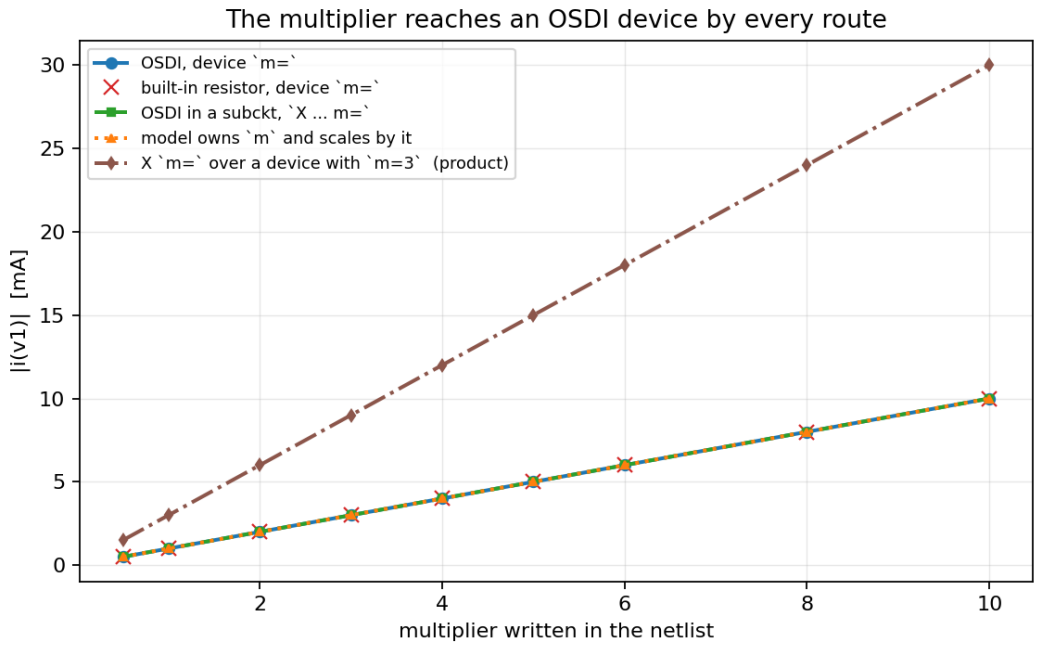
`has_m` suppresses *that alias only*. So when a model declares its own `m`, what it takes is the netlist keyword — `$mfactor` still exists, still defaults to 1, and is still reachable as `_mfactor`. The two are independent and they multiply:

netlist	model without its own `m`	model that declares and uses `m`
(nothing)	1× (`$mfactor`=1)	1× (`$mfactor`=1, m=1)
`m=3`	3× (`$mfactor`=3)	3× (`$mfactor`=1, m=3)
`_mfactor=2`	2× (`$mfactor`=2)	2× (`$mfactor`=2, m=1)
`m=3 _mfactor=2`	2× — one slot, last write wins	6× — both apply

The 2× in the bottom-left cell is a single parameter written twice; Enhancement-395's duplicate-write check reports it (*"'m' and '_mfactor' are the same parameter"*). The 6× is the one to watch: owning `m` does not disable `$mfactor`.

`integer m` behaves identically — `has_m` keys on the name only — except that `m=2.5` rounds to 3. Declare it `real` if fractional multiplicity matters; a plain model takes `m=2.5` as exactly 2.5×, because `$mfactor` is real.

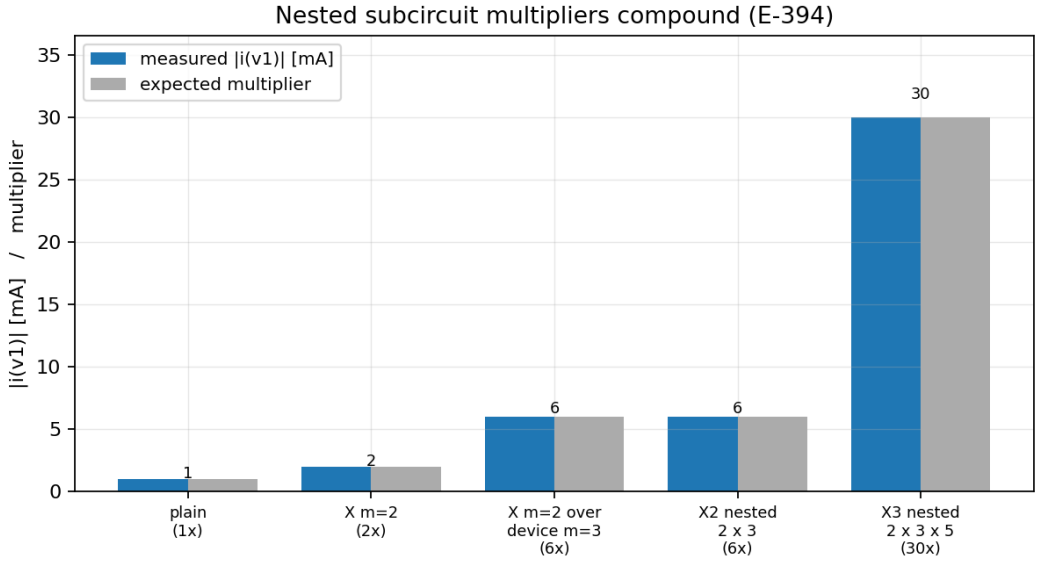
Every route delivers the same multiplier



Four routes lie exactly on top of one another — the OSDI device's own `m=`, the same `m=` on a built-in resistor, `X ... m=` on an enclosing subcircuit, and a model that owns `m` and scales by it. The fifth line is `X ... m=` applied over a device that already carries `m=3`: the two compound, giving 3× the rest.

Nesting compounds

Before Enhancement-394 the subcircuit multiplier never reached an OSDI device at all (1× in every analysis), and nested multipliers did not compound even for built-in devices — only the outermost `m=` survived. Both are fixed:



The multiplier applies in every analysis, not just DC — the regression suite pins AC and transient at m = 1, 3 and 7 as well.

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5. When the model declares one of the names

The single rule: if the Verilog-A declares it, the model owns it. ngspice hands over the netlist value and applies nothing of its own — for all four names.

```
// the CMC convention: declare `m` AND scale by it
(*type="instance"*) parameter real m = 1.0;
analog I(p,n) <+ m * V(p,n) * g;
```

With that model, `X1 a 0 sub m=3` gives exactly 3× and `$mfactor` stays 1 — there is no double application in either direction. The same model that declares `m` and then *ignores* it gets 1×, and the multiplier is lost. That is the model's bug, not the simulator's, and it is worth stating plainly because it is the trap that produces a plausible-looking false alarm: a probe module written to test the multiplier, declaring `m` without using it, looks exactly like "the subcircuit multiplier is silently defeated". It is not.

The same applies to temperature. A model declaring `dtemp` receives the netlist value in its own parameter and must add it itself:

```
(*type="instance"*) parameter real dtemp = 0.0;
analog begin
  Teff = $temperature + dtemp;   // ngspice did NOT add it
  ...
end
```

`$temperature` stays at the ambient in that case; `dtemp=10` and `dt=10` both deliver 10 to the model's parameter, and the effective temperature is whatever the model makes of it.

Why this is not reported as a collision

`dtemp` is a conventional CMC instance parameter — PSP 103/104, MEXTRAM 504/505, VBIC, BSIMBULK/CMG/IMG/SOI, HiSIM 2/HV/SOI/SOTB, L-UTSOI, EKV, MVSG, ASM-HEMT, JUNCAP200 and r2/r3_cmc all declare it. Because the loader *routes* its built-in to the model's parameter, both `IFparm` entries carry the same id and nothing is unreachable. A keyword-only collision check flagged all of them — Enhancement-396 made the check compare ids, which is what distinguishes a deliberate routing from a genuine `GAIN`/`gain` clash where one value really does become unreachable.

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6. Subcircuits

- **`.temp`** propagates into subcircuits and through nesting — a device three levels down sees the ambient.
- **`X1 ... temp=75`** does not work — and it does not work for a built-in device either. An `X` line binds subcircuit parameters, not device parameters. This is core ngspice semantics, not an OSDI shortcoming.
- The working idiom is a subcircuit parameter forwarded to the device:

```
.subckt s p n dtemp=0
N1 p n mm dtemp={dtemp}
.ends
X1 a 0 s dtemp=50        * -> $temperature = 350.15 K
```

- The multiplier is the exception: `X1 a 0 s m=3` *is* meaningful, because ngspice's expansion appends `m={m}` to each device line inside the subcircuit. That is what makes the multiplier compound across nesting levels.

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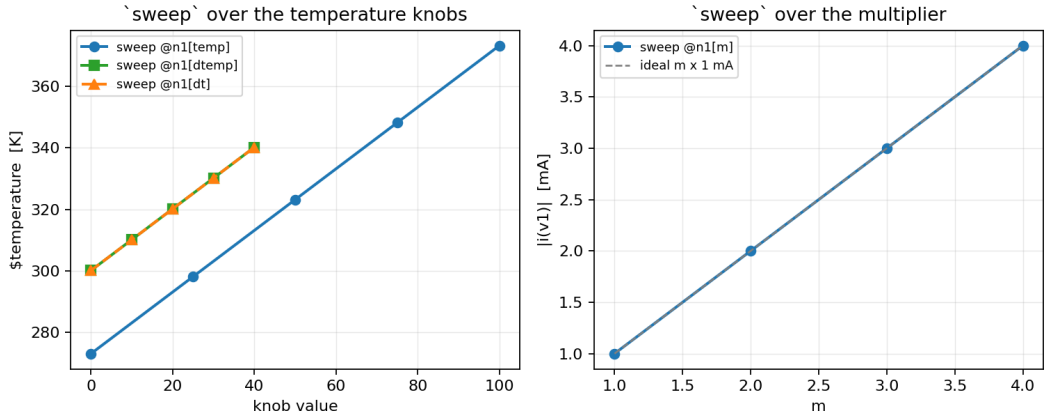
7. Sweeping these knobs

The `sweep` command (Enhancement-146) steps any knob and records an output. All four work directly:

```
sweep @n1[temp]  0 100 25 -analysis op -output tdev=@n1[tdev]
sweep @n1[dtemp] 0 40 10  -analysis op -output tdev=@n1[tdev]
sweep @n1[m]     1 4 1    -analysis op -output ii=i(v1)
```



Two practical notes:

- Do not give the output the same name as the knob. ngspice lowercases identifiers, so `sweep TT ... -output tt=...` collides and `print tt` shows the sweep *scale* instead of the recorded output.
- Sweeping `temp`, `dtemp` or `dt` leaves a trailing `Error: no such parameter temp.` after the run completes. The swept data is complete and correct — the message comes from §8.

If you would rather drive them symbolically, sweep a `.param` that feeds the instance line; this works for all four:

```
.param TSET=27
N1 a 0 mm temp={TSET}
...
sweep TSET 0 100 25 -analysis op -output tdev=@n1[tdev]
```

---

8. The one asymmetry that remains

ngspice registers its own `temp`, `dtemp` and `dt` entries as **`IF_SET`** without **`IF_ASK`**:

```
dst[0] = (IFparm){"dt",    (int)entry->dt,   IF_REAL | IF_SET, ...};
dst[1] = (IFparm){"dtemp", (int)entry->dt,   IF_REAL | IF_SET, ...};
dst[0] = (IFparm){"temp",  (int)entry->temp, IF_REAL | IF_SET, ...};
```

whereas a model's own parameters get both. They are therefore write-only:

	set it	read it back
`@n1[m]`	yes	yes (`$mfactor` is a real descriptor parameter)
`@n1[temp]`, `@n1[dtemp]`, `@n1[dt]`	yes	no — *"no such parameter temp."*
the same names on a built-in	yes	yes — `@r1[temp]` → 27, `@r1[dtemp]` → 0

So `show n1` on a model that declares none of them lists only `m`, and `print @n1[temp]` fails. This is why the three look absent unless the Verilog-A declares them — declaring one turns it into a model parameter, which carries `IF_ASK | IF_SET`.

The practical cost is small: a cosmetic trailing message after a sweep, and no read-back through `print`/`show`. Making them askable is contained but not a one-line change, because the synthetic ids overlap the range `OSDIask` uses for Enhancement-394's terminal currents — `entry->dt` is exactly `descr->num_params + descr->num_opvars`, which is also the terminal-current base. `OSDIask` would have to test the `dt`/`temp` ids first, mirroring what `OSDIparam` already does on the set side. Adding `IF_ASK` without that would make `@n1[temp]` return a terminal current: a wrong number instead of an error, which is worse.

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9. What is pinned, and where

`examples/instknobs_examples/verify_instknobs.py` — 86 checks, in the regression suite. It covers the multiplier by every route (device, subcircuit, nested, compounded, AC and transient, fractional, zero, 1000), temperature by every route against the built-in thermometer, `$vt` by the ratio test, the `m`/`$mfactor` independence including the 6× case, the integer rounding, subcircuit propagation and forwarding, and the model-owns-it rule in both directions.

Related notes and enhancements:

- Enhancement-394 — the subcircuit multiplier and the Celsius→Kelvin conversion
- Enhancement-395 — the duplicate-write check that reports `m` and `_mfactor` set together
- Enhancement-396 — the id-aware collision check, and the withdrawn `m` finding
- Enhancement-146 — the `sweep` command
- ngspice_osdi_bypass.md — the OSDI evaluation path

Reproducing the figures

```
python3 docs/internals/ngspice_internals/make_temperature_figs.py
```

Every figure in this note is produced by that script from simulations run at build time — the Verilog-A probe is compiled with `openvaf-r`, driven through the netlist forms described above, and plotted against the closed form or against a built-in device in the same deck. Nothing here is sketched.